

Math/Stat 452: Mathematical Statistics

Instructor: Benjamin Ridenhour

Spring, 2021

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Office Hours: MWF 9:30–10:20am
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Web: <https://bit.ly/2scoPgS>
Class Hours: MWF 8:30–9:20am
Class Room: TLC 030

Course Description

In this course, we will focus on understanding the interplay between probability, statistics, and modeling. In many regards, probability and statistics are complementary branches of applied mathematics. Probability expresses the chance of observing an outcome given a model (e.g., a distribution); conversely, statistics asks whether a given set of observations are consistent with a parameterization of a chosen mathematical model given an assumed probability distribution. More specifically, in statistics we often seek to find a range of parameter values in which we are confident or to distinguish between competing models with some degree of certainty. Likewise, “inverse problems” in mathematics attempt to find valid parameters given a generative model and observed data, which is simply another way to discuss the practice of statistical model fitting. This course will help you understand and implement the tools required to build complex models and fit them to real-world data.

Course Structure

Our class is intended to be an in-person class, and, as such, content delivery will be done via a live lecture. Per the University of Idaho administration:

Faculty must deliver courses according to the format listed in the course schedule. Students register and expect that their courses will be held in certain ways. Faculty should encourage students to attend classes in person whenever reasonable.

Obviously, given the circumstances of the COVID-19 pandemic, students will not always be able to attend class. **See the Healthy Vandals Policies at the end of this document for rules regarding attending class.** That being said, lectures will be broadcast via Zoom, recorded, and posted. Interaction with the Zoom audience will be minimal, therefore, I recommend you attend if at all possible. Similarly, evaluation (exams, quizzes) are being done in-person (see below).

Coursework will be as follows. Suggested homework problems will be given for each section of the book. Problem sets will be posted to BbLearn. Answers to the suggested problems can

be found in the back of the book. Friday class periods will be used for quizzes. The graded portion of the class will be your performance on three equally-weighted, non-cumulative exams, in addition to the Friday quizzes. I expect to give 9 quizzes of which you can drop one (so 8 quizzes count toward your grade). Quizzes will cumulatively count as one exam. In other words, each evaluated facet of class is worth 25% of your overall grade. Chapters of the textbook are fairly short. Exams will roughly cover about 9 chapters. *You will be allowed to have a page of notes during each exam, but no calculators are allowed!* **If you cannot be in attendance for a quiz or exam, arrangements need to be made with me prior to the event.**

Required Materials

- *Probability and Mathematical Statistics*. By MC Meyer. Society for Industrial and Applied Mathematics: Philadelphia.

Prerequisites/Corequisites

Prerequisites: Math/Stat 451

Topics By Week (*Approximate*)

1. Wks. 1–2: Introduction and review of distributions (discrete, continuous, multivariate); *No class MLK day 18 Jan.*
2. Wk 3: Chs. 42 (Parameter Estimation), 43 (MLE)
3. Wk 4: Chs. 43 (MLE cont.), 44 (MM; W&F)
4. Wk 5: Review, **Test 1**, Ch. 46 (Consistency)
5. Wk 6: Chs. 50 (CIs), 51 (Bootstrapping); *No class President's day 15 Feb.*
6. Wk 7: Chs. 52 (Information), 53 (Sufficient Statistics), 54 (UMVUE)
7. Wk 8: Ch. 55 (Exponential Family), Numerical methods, Review
8. Wk 9: **Test 2**, Hypothesis testing; *No class 12 Mar.*
9. Wk 10: *Spring Break*
10. Wk 11: Chs. 34 (Sampling a Normal), 35 (Hypothesis Testing), 36 (Hypothesis Testing)
11. Wk 12: Chs. 56 (Size and Power), 57 (Neyman-Pearson Lemma), 58 (LRT)
12. Wk 13: Chs. 58 (LRT cont.), CIs and testing, Review
13. Wk 14: **Test 3**, Ch. 59 (χ^2 -tests; W&F)
14. Wk 15: Models relying on normal distributions (M&W), Ch. 60 (ANOVA)

15. Wk 16: Chs. 45 (Bayesian Estimation), 49 (Credible Intervals; W&F)
16. Wk 17: Model selection, Hierarchical models, MCMC

Zoom Info

Meeting ID: 848 7841 1736

Passcode: 246270

Course Objectives/Learning Outcomes

Successful students will:

1. Understand the theory of statistics sufficiently to solve most problems appearing in standard textbooks for this course.

Final Exam Date

11 May (Tuesday), 8:00–10:00am

Grading Policy

A typical grading scale will be used with cutoffs every 10% (i.e., $\geq 90\%$ = A, $(90\%, 80\%]$ = B, ...). I reserve the right to curve the scale dependent on overall class scores at the end of the semester. Any curve will only ever make it easier to obtain a certain letter grade. The grade will count the assessments using the following proportions:

- 25% of your grade will be determined by each of the 3 exams.
- 25% of your grade will be determined by your quiz scores.
 - Calculated using the best 8 out of 9 quizzes.